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Experimental analysis of the effects of mesh geometries and electrowetting on fog water harvesting efficiency

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ABSTRACT

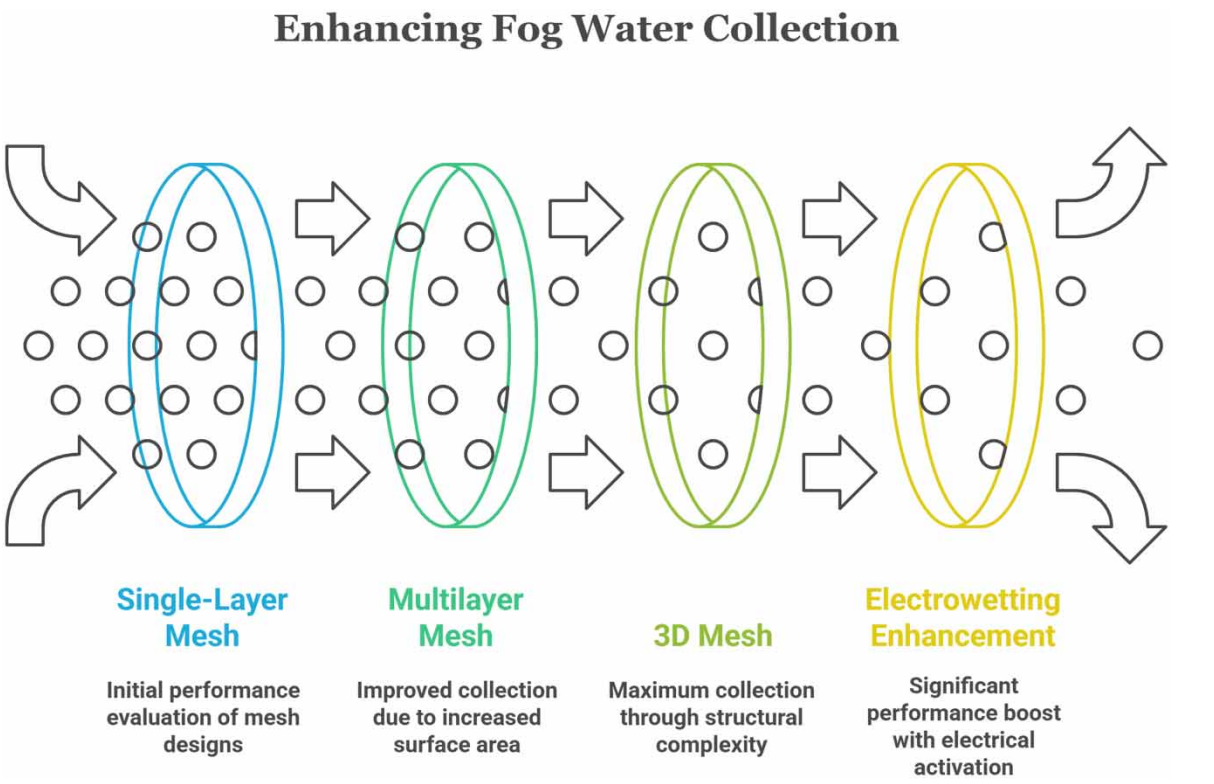
This investigation employed laboratory-controlled experiments to evaluate fog harvesting performance across multiple mesh configurations, integrating electrowetting techniques for enhanced fog water collection efficiency. In single-layer mesh, M1 (diamond shape) achieved optimal performance with 10.07 % collection efficiency and 39.21 ml of fog water yield. Multilayer configurations demonstrated superior performance, with M6 (M1 and M2) achieving 54.23 ml of collection attributed to increased surface area and optimized fiber architecture promoting hydrophilic interactions. The 3D mesh geometry yielded maximum fog water collection (99.43 ml), demonstrating the significance of structural complexity in fog capture mechanisms. Electrowetting led to substantial performance enhancement: mesh shown in Figure 8(a) (diamond) demonstrated voltage-dependent collection rates of 238 ml (24 V), 231.5 ml (12 V), versus 61.5 ml baseline. Similarly, mesh in Figure 8(b) (honeycomb) recorded 177.5 ml (24 V), 176 ml (12 V), compared to 50 ml without electrical activation. The hierarchical mesh design with active electrowetting created synergistic effects leading to enhanced fog water harvesting. The voltage-driven enhancement mechanism has significant potential for atmospheric water generation in water-scarce regions. Future development requires field validation, development of energy-efficient electrowetting protocols, durable hydrophobic coatings, and computational modeling for performance optimization of fog water harvesting in diverse environmental conditions.

Key words: collection efficiency, 3D mesh, fog, multilayer, water harvesting

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GRAPHICAL ABSTRACT



1. INTRODUCTION

Access to sufficient and sustainable water resources is one of the most vital factors in enhancing household income and promoting economic growth. The availability of drinkable water, necessary for human survival and well-being, is one of the natural resources most at risk from severe pressure. As climate change intensifies, global water demand continues to increase, and the projected water scarcity is expected to affect 6 billion people by 2050 (UNESCO 2018). Food security, climate change, environmental quality, and population poverty are all associated with freshwater scarcity, which has been considered as one of the major global issues (Mekonnen & Hoekstra 2016; Koop *et al.* 2022). The search for alternate water sources is imperative at this point and a key component of comprehensive water resource management and planning. They are effective means of reducing water scarcity and preventing the depletion of resources. There is thus a considerable need to formulate innovative technologies to address the issues confronted by remote regions or areas where the economic conditions inhibit the feasibility of a centralized treatment and distribution system (LaPotin *et al.* 2019). In places where economies of scale do not support centralized treatment and distribution networks, or to address the demands of rural communities, decentralized water production is a crucial tactic. Potential alternative water resources that are both renewable and economically viable must be investigated in these areas. In these regions, an alternative water source is fog present in the atmosphere, which can be harvested using novel techniques. Atmospheric water harvesting (AWH) provides an innovative solution for the decentralized production of potable water, particularly in high altitude hilly areas with limited access to liquid water. The atmosphere, a vast regeneration reservoir, holds roughly 12,900 km^3 of water, which is sufficient to partially supply the water requirements for industrial, agricultural, and residential uses (Tashtoush & Alshoubaki 2023). Fog harvesting offers a promising solution to the challenge of water scarcity, particularly in hilly regions where conventional water sources are limited. Recently, the implementation of such harvesting technology can be seen in different places around the globe. Additionally, the growth of these initiatives in Yemen, Nepal, and Ecuador enhances their credibility as a globally applicable solution (Urquidi & Campechano 2023). Although it is impractical for humans to rely only on fog drip (Klemm *et al.* 2012; Domen *et al.* 2014), fog water harvesting can be a substantial source of water in a variety of settings and uses.

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Typically, fog water is collected on a passive vertically raised collecting surface (mesh). According to Fischer & Still (2007), fog collectors can be roughly divided into two groups: active fog collectors and passive fog collectors. The mesh must have enough surface area for fog droplets to hit and adhere to while also permitting water droplets with sizes between 1 and 50 μm to flow through it to gather fog efficiently. As the velocity of the wind and/or the dimensions of the droplets increase in size, the resulting enhanced inertia of the droplets will improve their likelihood of contacting the mesh fibers. As the droplet size gets bigger, they flow under the effect of gravity. The design and material of the fog collectors used have a significant impact on the efficiency of fog harvesting systems, affecting how well they retain moisture and absorb droplets. The most common Standard fog water collectors (FWCs) are comprised of polymer meshes, particularly polypropylene (PP) and polyethylene (PE), attached to metallic frameworks aligned at right angles to the moist wind (Fernandez *et al.* 2018). The materials used and climatic factors of the region, like air humidity, wind speed, and fog composition, all significantly impact the effectiveness of FWCs (Domen *et al.* 2014). The standard FWCs can collect from 3 to 75 $L{\cdot}m^{-2}$ per day (Klemm *et al.* 2012).

Recently, the examination of impinging droplets on hydrophobic substrates has attracted considerable attention within the engineering discipline (Ruan *et al.* 2013; Jin *et al.* 2017). For a long period, inquiries into droplet impact phenomena have been carried out to scrutinize the intricacies of fluid dynamics along with the attributes of material surfaces, paying special attention to super hydrophobic interfaces. When a droplet collides with a substrate, it can either rebound, disperse and withdraw, trigger splashing, or settle on the substrate (Wu *et al.* 2015), subject to the prevailing environmental factors such as droplet diameter, viscosity, surface tension, and impact velocity from the liquid side; and surface roughness, wettability, surface shape, and surface temperature from the surface side (Lee & Kim 2014). In effect, a prompt rebound from the droplet causes a marked decline in the mass and energy interaction between the liquid drop and the hydrophobic surfaces. This phenomenon is contingent upon two critical parameters: the duration of contact and the surface area of contact. The investigation of droplet interactions with diverse geometrical configurations and surface textures was conducted employing molecular dynamics and Lattice–Boltzmann methodologies (Zhang *et al.* 2014). The conclusions drawn from the simulations were coherent at lower impact velocities, but discrepancies became evident at higher impact velocities. A considerable number of research studies have aimed at promoting a quicker separation of the impacting droplet from the substrate to decrease the length of contact (Lv *et al.* 2016; Patil *et al.* 2016).

Recent advancements in the field of materials science have also led to the development of innovative mesh structures that enhance the efficiency of water collection in fog harvesters by mimicking natural biological configurations and optimizing surface wettability characteristics (Gou & Guo 2019; Yi *et al.* 2019). Many other studies on improving the water collecting efficiency in several conditions of the laboratory have also been proposed (Park *et al.* 2013, 2014; Rajaram *et al.* 2016; Zhang & Wang 2016).The effectiveness of the water collection can be improved, for example, by modifying fiber morphology, incorporating steel harps, employing biomimetic materials, and integrating electrospun meshes that combine hydrophobic and hydrophilic polymers (Liang *et al.* 2017; Shi *et al.* 2018; Bhushan 2019; Knapczyk-Korczak *et al.* 2019). However, these techniques, intended to increase the effectiveness of water capture and harvesting, depend on sophisticated materials or intricate surface architectures that call for comprehensive microfabrication processes or methods and function in a passive way.

Other investigations explored the phenomenon of droplet impact on both smooth and textured surfaces. For example, Fink *et al.* (2018) executed an empirical analysis, noting that droplet deposition occurred on flat substrates while textured substrates resulted in bouncing followed by subsequent deposition. The maximum spreading diameter exhibits a direct correlation with the droplet diameter and impact velocity, while displaying an inverse relationship with viscosity and surface tension (Lee & Kim 2014). Rougher substrates, particularly at elevated impact velocities, are prone to inducing immediate splashing in contrast to their smoother counterparts (Li *et al.* 2021). A water contact angle of more than 150° induces water repellence, resulting in a super hydrophobic surface (Zhu 2018). Some investigations in this category indicate that the surface morphology in submillimetre scales plays a vital role in reducing the contact time of impact droplets by breaking the symmetry mode of the spreading process (Gauthier *et al.* 2015; Shen *et al.* 2017). Kim *et al.* (2020) examined the dynamics of droplet volume fluctuation, revealing the significant influence of gravity on droplet tip motion. Many other scholars have also worked on the dynamics of droplet wetting on textured surfaces (Omeje & Itina 2022; Antonov *et al.* 2023; Wang *et al.* 2023; Josyula *et al.* 2024). Among the computational methods, the Lattice–Boltzmann method (LBM) has become a promising numerical technique for studying multiphase flows with

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complex interfacial dynamics (Ezzatneshan & Khosroabadi 2021; Sharoonizadeh *et al.* 2022; Shahabi *et al.* 2023).

Compared with other studies to improve the collection efficiency, one such method is the use of an electric field (Kim & Kaviany 2007; Baratian *et al.* 2018) to alter the wetting characteristics due to surface tension for improving the collection efficiency. Also referred to as the electrowetting concept, it has demonstrated remarkable effectiveness in achieving contact angle fluctuations of several tens of degrees, which are frequently accomplished. The electrowetting phenomenon can be described in terms of forces resulting from an applied electric field (Cheng & Chen 2010). The electric field tends to pull the edges of the droplet down onto the electrode, which causes an increase in the droplet contact area and a (macroscale) decrease in the contact angle. Therefore, electrowetting can be defined as altering a surface's wetting characteristics, which are normally hydrophobic. Pollack *et al.* (2002) utilized electrowetting to control microdroplets (3 nl–μl) on a flat electrode setup, achieving swift movement (>0.1 m/s) without the need for permanent channels. A study by Kuiper & Hendriks (2004) showcased electrowetting-based adjustable liquid lenses by modifying the meniscus curvature between immiscible liquids, allowing for a compact achromatic camera module. Researchers such as Yan (2019) have explored passive methods using advanced materials, but electrowetting-on-dielectric (EWOD) has emerged as an active approach due to its fast, reversible, and reliable surface wettability control. Another study by Wikramanayake & Bahadur (2019) demonstrates that electrowetting enhances condensation heat transfer by promoting droplet coalescence, growth, and roll-off. Higher voltage and AC electric fields improve droplet dynamics, leading to a more efficient condensation process. The findings show over 30% enhancement in condensation rates, proving electrowetting as a promising method for improving heat transfer efficiency. These investigations underscore the potential of the electrowetting concept in the realms of advanced microfluidics manipulation of surface and enhancing the water collection of fog harvesting.

Recent trends in fog harvesting technology reflect an increasing integration of advanced materials and innovative designs. For instance, the use of Janus meshes has emerged as a promising strategy to enhance drainage and minimize clogging, a common issue with traditional mesh designs (Lee *et al.* 2022). Additionally, three-dimensional (3D) structures inspired by origami and biological models have shown great potential in maximizing capture surface area, allowing more effective condensation and collection of droplets (Nguyen *et al.* 2021; Peng *et al.* 2023). The inclusion of biomimetic approaches, where designs are influenced by natural systems such as the capabilities of cacti and beetles, further supports the exploration of novel geometries that enhance water capture efficiency (Yi *et al.* 2019).

Overall, the comparative analysis of mesh designs for fog collectors and the manipulation of the miniature wetting surface of fluid will reveal a diverse array of innovations aimed at improving water collection efficiency. This research aims at an integral study of the above mentioned passive and active approaches for potential improvements in collection rates under laboratory conditions. This experimental investigation will explore how various mesh geometries and the electrowetting concept affects the efficiency of fog water harvesting by modifying the hydrophilic nature of fluid. The findings could prove critical not only for enhancing fog harvesting methodologies but also for promoting sustainable water management practices.

2. MATERIALS AND METHODS

2.1. Fog collection experiment

2.1.1. Fabrication of Fog collector designs

The laboratory experimental setup fabricated for fog harvesting mechanism, with the main parts of the system, is shown in Figure 1. The different components shown in Figure 2 consist of (a) commercial humidifier, (b) fog transfer line, (c) humidity chamber with mesh stand, (d) water collector, (e) wind anemometer, and (f) humidity meter. A humidity chamber of about 40 × 35 cm (L × W) was made by using fiber plexiglass sheet. Different designs of mesh (Figure 3) in the mesh stand were placed inside the humidity chamber and tested to observe the water collection efficiency. A square shape mesh collector of about 24 × 27 cm (L × W) with eight different mesh materials and geometries was prepared and tested. The experiment was tested with different mesh collectors to see the difference in water collecting efficiencies. Eight different types of mesh collectors as shown in Figure 3 were used in this study: M1 (diamond shape), M2 (square), M3 (shade net), M4 (honeycomb), M5 (vertical array of aluminum wire), M6 (overlapped M1 and M2), M7 (overlapped M3 and M4 aluminum), and M8 (overlapped M1 and M3).

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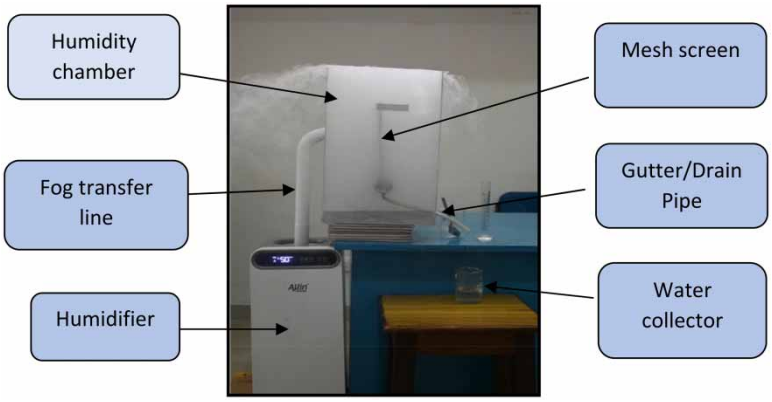


Figure 1 | Experimental setup of study of different mesh water collection efficiencies.



Figure 2 | Materials used for the first experiment.

2.2. Experimental methodology

2.2.1. Experiment on different geometry mesh designs

Initially, the water collection efficiency of eight distinct passive mesh structure designs was examined. A humidifier was used to generate fog, which can effectively control the indoor humidity of interiors within 40–90% and

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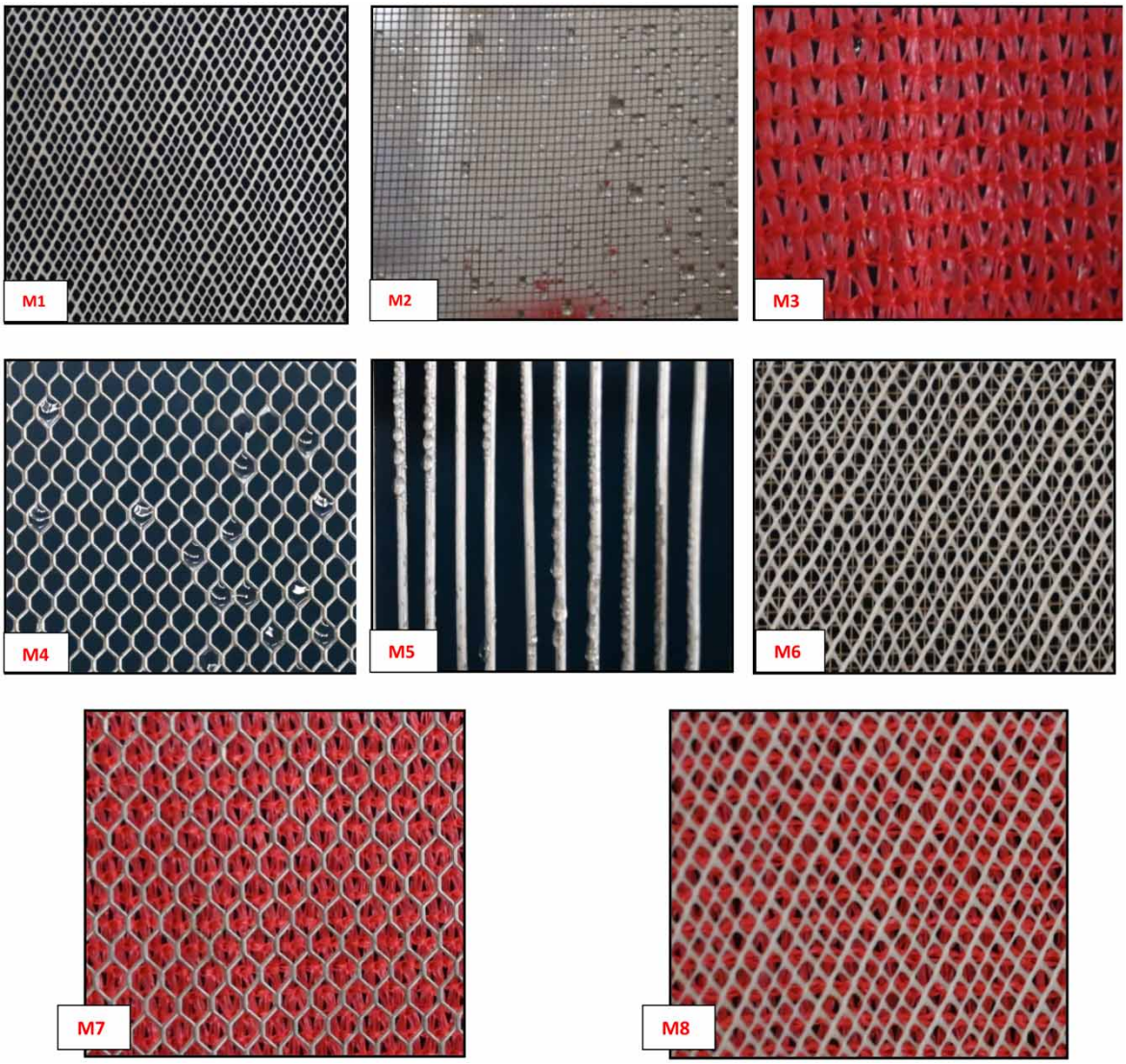


Figure 3 | Different mesh designs for performance evaluation.

can have custom humidity levels in different environmental conditions, a plastic pipe for the passage of mist flow towards the humidity chamber where the various designs of mesh collector were placed for water collection efficiency test. A relative humidity of 95% was maintained inside the chamber for this study. The fog collection experiment was performed on 8 different mesh designs with various shade coefficients. The mesh specifications and other properties of the mesh materials (size, shape, shade coefficients, etc.) are also given in Table 1. For optimal impaction, the pipe was intended to produce the fog in the center of the screen. The fog humidifier was run initially for around 20 min to maintain the required humidity (95%) inside the humidity chamber. After the humidity reaches around 95%, condensation occurs; thereafter, the deposition of small droplets starts on the collecting surface of the mesh structure. Each mesh structure was tested for a total running time of 1 h, where the observation (water droplets collected) was noted down at a uniform interval of 20 min.

2.3. Mathematical model

The assessment of fog water collection efficiency, including aerodynamic and deposition and drainage efficiency, is contingent upon various equations and foundational assumptions. Central to these are the Stokes number and the drag coefficient.

In this work, the overall collection efficiency (η) was determined for 5 different single-layer mesh structures (M1–M5). The approach for the calculation of η is given in the following sections:

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Table 1 | Mesh specifications and other properties

Collector	Mesh specification		Distance b/w two wires (mm)	Shade coefficients	Liquid Water Content (LWC) (gm/m^3)
	Mesh screen size (L × W)cm	Shape of pore space			
M1 Fiber (white)	(24 × 27)	Rhombus	0.2	0.55	0.5
M2 Fiber(black)	(24 × 27)	Square	0.15	0.53	0.5
M3 Nylon	(25 × 28)	Inverted triangle	0.1	0.5956	0.5
M4 Aluminum	(24.5 × 27.5)	Honeycomb	0.5	0.2	0.5
M5	(24 × 27)	Vertical array of wires	0.4	0.2667	0.5

a) Overall collection efficiency was calculated by the formula given below (de Dios Rivera 2011)

$$\eta = \eta_a \times \eta_{\text{dep}} \times \eta_{\text{dr}} \tag{1}$$

where η_{a} is the aerodynamic efficiency, η_{dep} is the deposition efficiency, and η_{dr} is the drainage efficiency. Aerodynamic efficiency (η_{a}) denotes the water droplet fraction in the fog that may collide with the mesh. The fraction of actual fog droplet that impinges on the mesh fibers and which get deposited is represented by the capture efficiency (η_{cap}). The drainage efficiency (η_{dr}) represents the fraction of the fog water that travels to the gutter after colliding with the mesh.

2.3.1. Aerodynamic efficiency

Aerodynamic efficiency of the collector is given by the following approximate expression, given by (de Dios Rivera 2011):

$$\eta_a = \left(\frac{\text{SC}}{1 + \sqrt{\frac{C_o}{C_d}}} \right) \tag{2}$$

where SC is the shade coefficient, which is the fraction of mesh area that is occupied by mesh wires; C_{d} is the drag coefficient of screen; C_{o} is the pressure drop coefficient. The drag coefficient (C_{d}) is a crucial parameter in determining the aerodynamic efficiency (η_{a}) of the collector, which represents the fraction of air passing through the mesh rather than around it. The aerodynamic efficiency equation also incorporates the shade coefficient (SC) and the pressure drop coefficient (C_{o}). The mesh SC represents the capturing capability of droplets since only part of the mesh can catch droplets, and it is defined as (de Dios Rivera 2011):

$$\text{SC} = 1 - \frac{A_{\text{op}}}{A_{\text{tot}}} \tag{3}$$

The proposed empirical formula for C_{o} for a wire mesh is given by (de Dios Rivera 2011):

$$C_o = 1.3\ \text{SC} \left(\frac{\text{SC}}{1 - \text{SC}} \right)^2 \tag{4}$$

In the study, the value of the drag coefficient (C_{d}) is generally taken as a constant value of 1.18, as suggested by previous research (Rajaram *et al.* 2016; Gandhidasan *et al.* 2018).

2.3.2. The deposition efficiency

The deposition efficiency depends on the dimensionless Stokes number and is given by (Park *et al.* 2013):

$$\eta_{\text{dep}} = \frac{\text{St}}{\text{St} + \frac{\pi}{2}} \tag{5}$$

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where the Stokes number

$$\mathrm{St} = \frac{\rho_a D g^2 V}{18\,\mu_a R} \tag{6}$$

with ρ_a (kgm^{-3}) being the density of air, D_g (m) is the fog droplet diameter, V(ms^{-2}) is the wind speed, μ_a(Nsm^{-2}) is the dynamic viscosity of the air, and R(μm) is the radius of mesh fiber. The Stokes number (St) denotes a dimensionless variable that analyses the kinetic relationship between particles and gas. It is defined as the ratio of the particle's characteristic relaxation time (t_0) to the flow's characteristic time (l_0/u_0). The particle diameter (dp), particle density (ρ_p), and gas dynamic viscosity (μ_g) are used to calculate the relaxation time (t_0). The combination of these two equations of relaxation time and the general stokes formula yields the final Stokes equations as given above in Equation (6). Deposition efficiency (η_{dep}), which is a component of overall collection efficiency, primarily depends on inertial impaction for the fog droplet sizes considered (1–50 μm) (Mahat *et al.* 2019). Mechanisms like Brownian diffusion and interception are considered negligible for these droplet sizes (Fernandez *et al.* 2018).

The drainage efficiency is estimated as,

$$\eta_{dr} = \frac{\eta_{dep}}{\eta_{imp}} \tag{7}$$

where η_{dep} and η_{imp} are the deposition and impaction efficiencies, respectively.

Efficiency due to impaction is given by Ritter *et al.* (2008):

$$\eta_{imp} = \frac{\mathrm{St}^2}{(\mathrm{St}+0.6)^2} \tag{8}$$

for $\mathrm{St} \geq 0.08$ and if $\mathrm{St} < 0.08$, then $\eta_{imp} = 0$

2.4. Experiment on electrowetting application

Electrowetting can be expressed as a modification of the wetting properties of a droplet surface (Figure 4), which is typically hydrophobic, using an applied electric field. Electrowetting is highly successful in contact angle variations of several tens of degrees. In this part, we go beyond the passive way of fog collection by use of a direct current (DC) electric field in an electrowetting configuration to manipulate the wetting characteristics of the droplet. The experimental setup from part (A) was modified with the following materials (Figure 5) as shown in Figure 6. A similar procedure was followed as in the previous experiment. Other than that, the fog produced from the exit end of the pipe was made to pass through an electric field before impacting the selected three-dimensional (3D) mesh design.

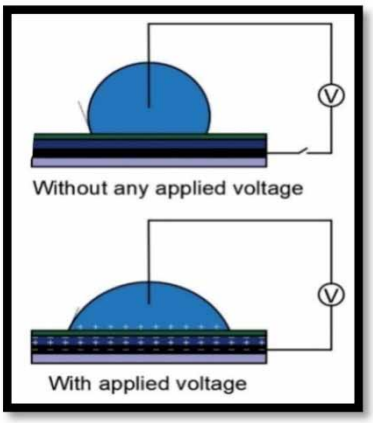


Figure 4 | Water droplet with and without electric field.

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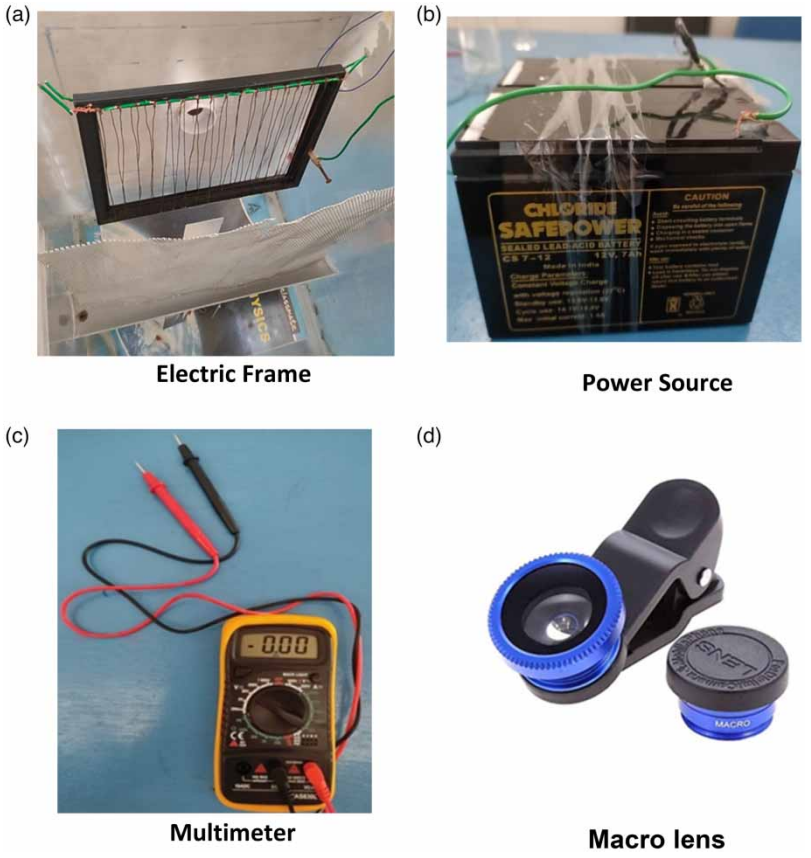


Figure 5 | Materials used in the electrowetting experiment.



Figure 6 | Electrowetting laboratory fog harvesting experiment setup.

2.4.1. Contact angle measurement

The contact angle is the angle formed by the intersection of the liquid–solid interface and the liquid–vapour interface. If the contact angle is low, then the wetting capability will be high and vice versa (Figure 7). To ensure that the electrowetting effect was present before testing, the contact angle of a single droplet was measured on the devices under different DC voltages of 0, 12, and 24 V. The fog produced was exposed to an electric field on its way before impacting. The fog droplets were ionized after passing through the electric field. The process was repeated for different applied voltages. The experiment was performed to compare the amount of fog water collected and the droplet area coverage between charged and uncharged samples, and an improvement in water capturing capability under different voltages of the electrowetting effect. Image of the fog water droplet

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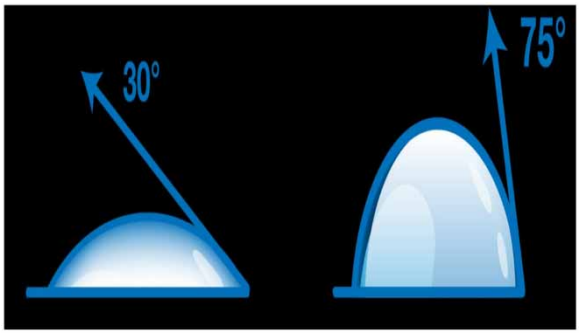


Figure 7 | Contact angle of water droplet (low contact angle – good, high contact angle – negligible wetting).

under different applied voltages (0, 12, and 24 V) was clicked using a macro lens. ImageJ software was used to measure the contact angle of water droplets.

3. RESULTS AND DISCUSSIONS

3.1. Mesh intercomparisons

The experiment was conducted for eight selected mesh designs (Figure 2) with different shade coefficients. The fog interception and collection capabilities of mesh structures were examined using a consistent fog/mist flow with an air flow rate of 0.05 m/s. The individual mesh structure frame was placed at 90° with the exit end of the fog pipe. The size of the deposited droplets ranged from 3 to 5 mm. The trial run intercomparison was done in two groups: First comparison was amongst the single-layer mesh designs (M1–M5), Second comparison was amongst the double layer mesh designs (M6–M8). The results of the mesh comparisons and experimental values of aerodynamic efficiency, deposition efficiency, overall efficiency, and the cumulative amount of fog water collected are shown in Table 2. It can be observed from the results that in the first group, M1 (diamond) shape mesh design, showed the highest overall efficiency and amount of fog water collection of 10.07% and 39.21 ml, respectively. M1 mesh is made up of smaller-diameter fiber wire (1 mm), which has led to an increased surface area for droplet adhesion, thereby increasing collection efficiency. The smaller diameter wire maximizes the density of fog collision sites and total surface area available for droplets to impact and adhere to, while maintaining the airflow. Another aspect is the promotion of faster droplet coalescence and runoff, which is due to reducing surface tension effects. Similar findings related to thinner diameter wires leading to increased collection efficiency were also reported (de Dios Rivera 2011; Park *et al.* 2013). Also, the SC of the design was worked out to be around 50%, due to which the aerodynamic efficiency and the overall efficiency were high. Additionally, M1 and M2 exhibit greater aerodynamic efficiency (0.28 and 0.27, respectively) than the other designs (0.3, 0.18, and 0.22 for M3, M4, and M5, respectively). This higher aerodynamic efficiency is influenced by the higher SC in the

Table 2 | Mesh intercomparisons and experimental values of overall water collection and cumulative amount

Mesh	Efficiency parameters						Overall η	Avg. water collected (in ml) at different time intervals (min.)			Avg. water collected in 1h (ml)
	Co	η_{Ia}	St	η_{dep}	η_{imp}	η_{dr}		20	20	20	
M1 (Rhomb us)	1.39	0.28	0.885	0.36	0.35	1.00	10.07	13.57	12.01	13.63	39.21
M2 (Square)	1.68	0.27	0.885	0.35	0.35	1.00	9.45	8.53	8.57	13.50	30.06
M3 (Inverted triangle)	0.65	0.3	0.177	0.10	0.05	1.00	3	5.20	5.53	5.57	16.3
M4 (Honeycomb)	0.02	0.18	0.885	0.36	0.35	1.00	6.48	6.83	6.93	5.63	19.4
M5 (Vertical array of wires)	0.05	0.22	0.5	0.36	0.35	1.00	7.92	7.13	7.27	6.90	21.3
M6 (DoubleLayeredM1&M2)								17.07	20.17	17.0	54.23
M7 (DoublelayeredM3&M4)								15.5	18.33	17.03	50.87
M8 (DoublelayeredM1&M3)								12.27	13.70	12.1	38.80
M9 (3DSolid Combo)								35.63	31.53	32.27	99.43

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range of 50–53% which was observed in both the designs. Similar results of relationship with aerodynamic efficiency and shad coefficients were also observed in a study by Mosa *et al.* (2024). In this study, the drainage efficiency (η_{dr}) was taken as 1. The stoke number (M1 and M2: 0.885) was also high for this design, which may have contributed to a higher overall collection efficiency as observed from Table 2. A higher Stokes number is a desirable characteristic for fog collection, as it directly translates to a greater probability of fog droplets colliding with and being captured by the collection mesh (Fernandez *et al.* 2018). M2, M5, and M4 followed by M3 mesh designs showed an overall efficiency and fog water collected value of (9.45%) 30.06 ml, (7.92%) 21.3 ml, (6.48%) 19.4 ml, and (3%) 16.3 ml, respectively.

A similar trend of results was observed for the above mentioned mesh design. The various shape of the mesh design, M1: diamond, M2: square, M3: commercial shade net, hexagonal and M5: vertical array have also influenced the collection rate of the fog droplets. The diamond pattern in M1 has delivered better performance among the different designs in this study. Similar findings was also reported by Lee *et al.* (2022) where they indicated that specific geometric configurations, such as diamond-holed Janus meshes, significantly improve water drainage along the surface, thereby optimizing fog harvesting yields.

The second case of the experiment was performed between the double layered mesh design, which is of the following combination as follows: M6 (M1 and M2), M7 (M3 and M4), and M8 (M1 and M3). M6 was an overlap between M1 and M2, resulting in the highest collection of fog water (54.23 ml). A collection time of 60 min was also followed in this trial. M1 and M2 showed a collection amount of 50.87 and 38.80 ml, respectively. M6 has outperformed with a percentage increase of 31.11% from the other two overlap designs, as can be seen from the Table 2. This is mostly due to the increase of surface area and the nature of wires (fiber), which showed a hydrophilic nature of water repellent for the M1 overlap design. The above two groups of mesh designs experiment were performed in a two-dimensional (2D) passive collecting surface area. Also, layered mesh design intercepts droplets at multiple stages. Ghosh *et al.* (2021) also reported a similar increase in efficiency by 20–50% of layered designs compared to single-layer meshes. Choi *et al.* (2023) also used a 3D printing assisted fabrication of hierarchically structured biomimetic surfaces with dual-wettability for water harvesting. In this study, the M9 mesh collector used represents a 3D design from a study by Yurembam *et al.* (2023). The design is a combination of two surface features: of rectangular face trapezoid and a semi-arch solid. It was observed that this 3D mesh showed the highest amount of water collected, of around 99.43 ml. The increase of about 153.58% was visible with the use of 3D mesh design over layered and single-layer mesh among the highest-performing designs.

3.2. Electrowetting experiment

The fog interception and collection capability of the two different mesh structures (Figure 8(a) and 8(b)) was examined under laboratory conditions. The individual mesh structure frame was placed at 90° with the protruded plastic pipe to determine the fog collecting efficiency. The model was run for 30 min, and the readings were noted. The experimental setup is shown in Figure 6. The wired frame electric field was placed parallel to the mesh structure. Copper wire was used to develop an electric field net and was placed in front of the mesh collector. The fog droplets get ionized after passing through the net. The experiment was replicated to ensure accuracy. The results of the runoff dynamics are given in Table 3. The experimental trials were done under three categories:

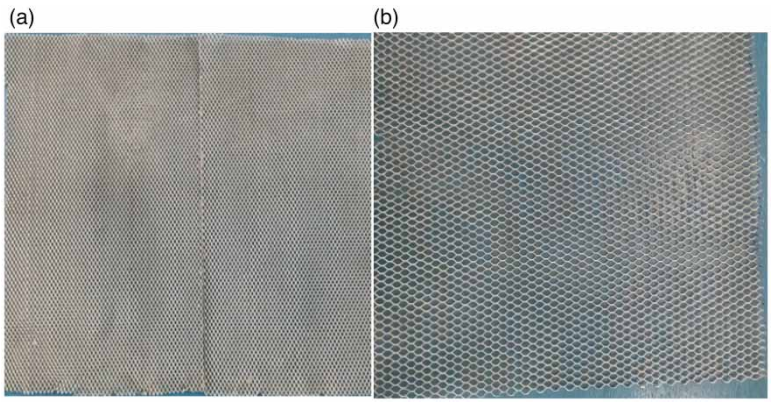


Figure 8 | (a) Diamond shape and (b) honeycomb shape mesh structure.

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Table 3 | Experimental results of fog water runoff dynamics of wired mesh structures

Sl. no.	Mesh	Electrical field	Water runoff dynamics over 30 min (ml)				Surface of contact (mm^2)	Water Collected (ml)
			First	Second	Third	Mean		
1	Figure 8(a)	Absent	17	19	25.5	20.5	0.215	61.5
		12 V	91.5	72	68	77.16	0.127	231.5
		24 V	80	76	82	79.3	0.307	238
2	Figure 8(b)	Absent	15	12	23	16.7	0.214	50
		12 V	60	61	55	58.7	0.452	176
		24 V	60	57	60.5	59.16	0.105	177.5

(a) without any voltage (absent), (b) under applied 12 V, (c) under applied 24 V. From Table 2, the highest amount of fog water collected for mesh (Figure 8(a)) was observed as 238 ml for a 30-min runoff dynamics and under an applied voltage of 24 V. The 12-V trial also showed a little lower collection of about 231.5 ml for the same period. The normal trial without any electrowetting induction at the same collection time was observed to be 61.5 ml. Among the three different trials, the case of applied voltage (12 and 24 V) showed an improved water collection volume of around 286.99%, compared to the first trial (without voltage). Likewise, a similar scenario was observed in the experiment conducted for the second mesh design (Figure 8(b)). Here, the highest amount of fog water collection was observed for the 24-V applied electric field, followed by 12 V and without electric field trials as 177.5, 176 and 50 ml, respectively. The overall percentage volume collection was also observed to be around 252%. It can also be observed that for the two different mesh designs (Figure 8(a), 8(b)), the amount of water collected for the 12- and 24-V ranges was almost similar.

Electric fields can induce variations in droplet shapes, support the process of merging, and promote detachment of droplets by changing surface tension and contact angles, particularly on conductive or non-conductive materials. The equilibrium contact angle and the general topology of a droplet on a surface are controlled by the balance of interfacial tensions between the liquid-gas phases, which in turn determines the wettability of a surface. In this study, the different mesh was all placed in vertically inclined positions, which induces angle hysteresis. The application of DC voltage results in a reduction of the critical angle of inclination required for the mobility of liquid water droplets. This reduction explains the differing behaviors of droplets and the resulting water accumulation at specific applied voltage levels (Kim *et al.* 2013). Additionally, the electric force caused an imbalance in the droplets' surface tension forces, which started the motion of the droplets. Due to its cheap cost and power consumption, this technique has the potential to enhance water harvesting systems by actively shedding droplets and causing collisions that increase droplet growth (Li & Mugele 2008). Similar studies also reported the improved droplet coalescence and growth, as well as drainage leading to larger droplets (Högnadóttir *et al.* 2020). This was also supported by the volume of water collected under different applied voltages (Table 3).

3.3. Contact angle measurement

The contact angle measurement was done under different DC voltages of 12, 24 V and also without an electric field. Measurements performed on the two meshes (Figure 8(a) and 8(b)) are shown in Figure 9(a) and 9(b). The trial period of the laboratory fog harvesting was run, and at the end of the 30 min, an image of the fog water droplet was taken using a macro lens. ImageJ software was used to measure the contact angle of water droplets clicked under the charged (12 and 24 V DC) and uncharged conditions. From the measurement (Figure 9(a)), it was found that the contact angle for droplets without an electric field:100.1°; with an electric field (12 V): 97.17°; and with an electric field (24 V): 90.9°. It was observed that the contact angle without an electric field (100.1°) is more than the contact angle of droplets with an electric field (12 V:97.17°), which therefore increases the shedding rate of droplets with the electric fieldpolarization and hence increases the efficiency of fog water collection. Wikramanayake & Bahadur (2019) also have demonstrated that electrowetting enhances condensation heat transfer by promoting droplet coalescence, growth, and roll-off. Similarly, in the measurement of mesh designing Figure 8(b), it was observed from Figure 9(b) that the contact angle for droplets without an electric field: 101.3°; with an electric field (12 V): 90.00°; and with an electric field (24 V): 94.71°. The decrease in the contact angle in both observations can be expressed as a modification of the wetting properties (Kuiper & Hendriks 2004) of a surface using an applied electric field.

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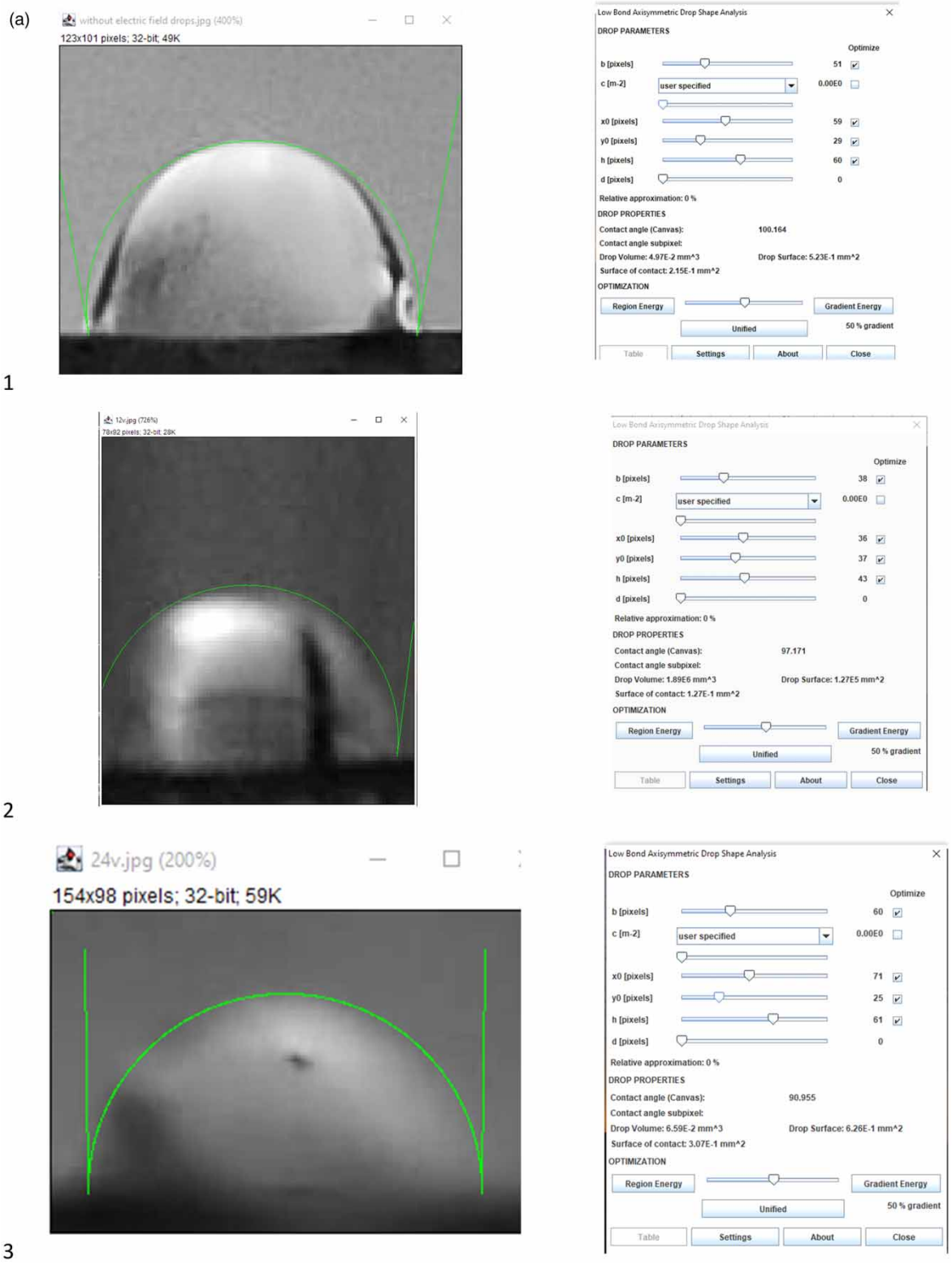


Figure 9 | (a) Experiment contact angle measurement: (1) without electric field, (2) with 12 V applied voltage, (3) with 24 V applied voltage. (b) A2 experiment contact angle measurement: (1) without electric field, (2) with 12 V applied voltage, (3) with 24 V applied voltage. (*continued.*).

Electrowetting enhances fog harvesting by actively modulating droplet wettability/contact angle. When voltage is applied, the contact angle θ of conductive droplets decreases (Lippmann 1875), promoting better surface wetting. The mist-filled air from the exit pipe was allowed to pass through an electric field before hitting the mesh

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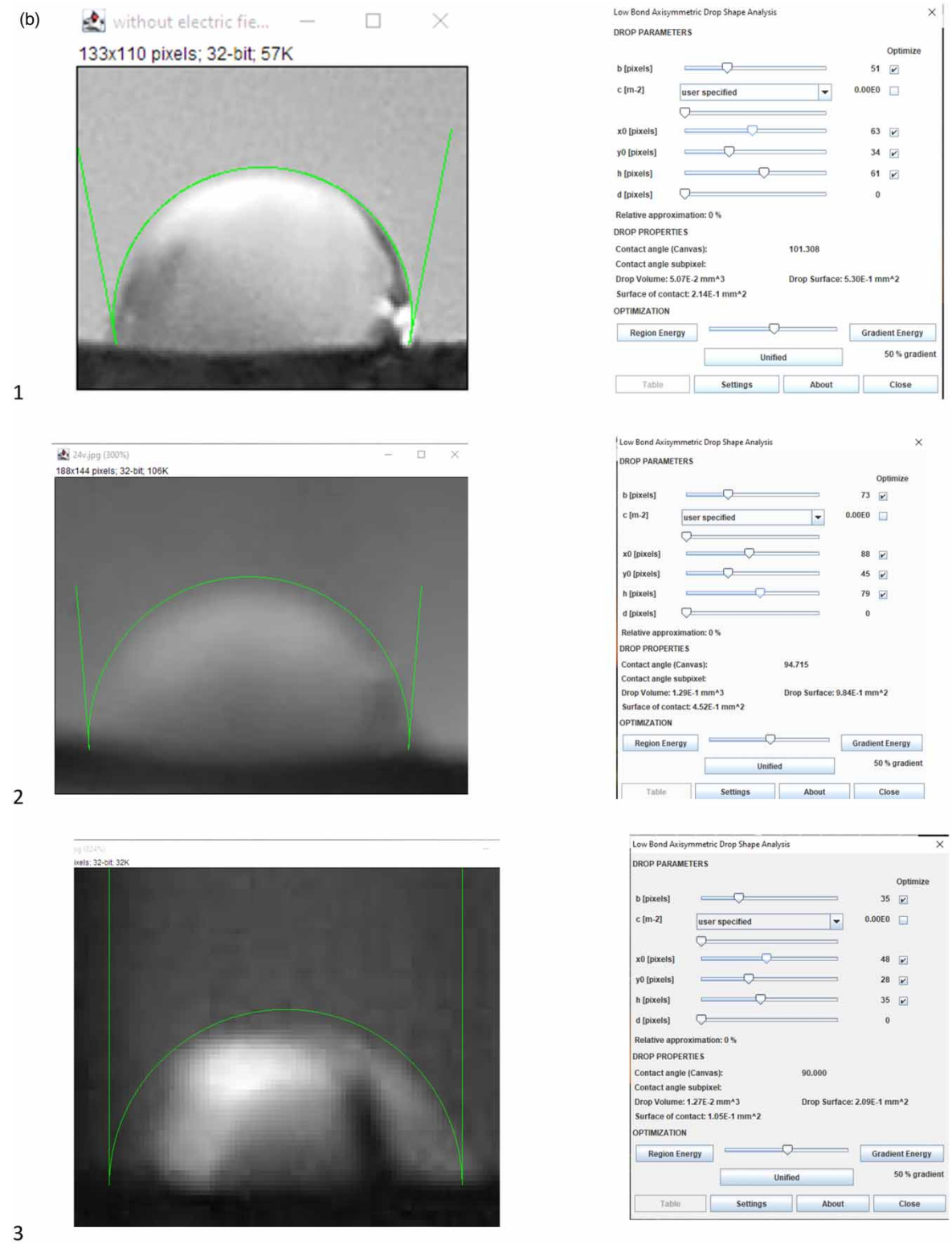


Figure 9 | Continued.

collector placed at a relatively short distance from the electric field. This improved adhesion reduces droplet bounce-off and encourages faster coalescence into larger droplets, aiding efficient drainage. Even before impact, fog droplets traversing through an electric field become polarized and slightly elongated, which can alter their effective contact angle on landing. This was visible from the contact angle changes as shown in Figure 9(a) and 9(b).

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In this experiment, the uncharged droplets are made to travel through a uniform electric field setup. During this process, the charges within the droplet redistribute. Positive charges accumulate on the side facing the negative electrode, whereas negative charges accumulate on the side facing the positive electrode. The droplet becomes polarized as a result of the electric dipole moment created by the charge separation. Upon impact, electrostatic forces at the contact line enhance spreading, through excessive voltage risks dielectric breakdown. Experimentally, contact angles in the first trial mesh (Figure 8(a)) have been reduced from 4.13° and 6.27° as stepwise reductions and a total reduction of 10.4° in the absence, 12 and 24 V applied cases. Likewise, for the second trial (Figure 8(b)), a stepwise reduction in contact angle from 6.55° and 4.75° with an overall reduction of 11.3° was observed. Both trials (Figure 9(a) and 9(b)) showed a decrease in contact angle, with the largest reduction occurring in the second step for Figure 9(a):6.27° and second step for Figure 9(b):6.55°. A study by Welters & Fokkink (1998) also demonstrated control over the droplet contact angle by applying potential between a water droplet on a hydrophobic coating and the underlying conductive substrate. The total reduction of the two trials in contact angle measurement from initial to final reading is 10.4° and 11.3°, respectively. The external electric field causes the droplet's surface charges to react, creating an outward electrostatic pressure that acts perpendicular to the droplet surface. The differential pressure induces droplet deformation and elongation, orienting its major axis along the electric field direction. These forces have the potential to surpass the force of surface tension, resulting in faster deformation and a greater accumulation of water under various applied voltages. By increasing droplet mobility, the electric force also encourages droplet coalescence. Because of gravity or other hydraulic forces, this may result in larger droplets that are easier to remove from a surface. The findings show that electrowetting-modified experiments achieve 9–12% lower contact angles during active collection, enhancing the water capture efficiency.

4. CONCLUSIONS

The laboratory experiment evaluating fog harvesting performance across various mesh designs, coupled with electrowetting trials, gave critical insights into various aspects of water collection efficiency. In Mesh intercomparisons, for the single-layer mesh 'M1' mesh outperformed other mesh designs with the highest overall efficiency and volume of water collection of 10.07% and 39.21 ml. For the multilayer mesh designs, M6 has outperformed the other two overlap designs with the highest water collection amount of 54.23 ml. This is mostly due to the increase of surface area, geometry of mesh and the nature of wires (fiber), which showed a hydrophilic nature of water repellent for the M1 overlap design. It was also observed that a 3D mesh showed the highest amount of water collected at 99.43 ml. Also for mesh Figure 8(a), the highest amount of water collected was observed as 238 ml for a 30-min runoff dynamics and under an applied voltage of 24 V. The amount collected for 12 V was about 231.5 ml as compared to 61.5 ml without any voltage exposure. Likewise, in the second mesh (Figure 8(b)), the highest amount of water collection was observed for the 24 V applied electric field, followed by 12 V and without electric field trials as 177.5, 176 and 50 ml, respectively. It can also be observed that for the two different mesh designs, the amount of water collected for the 12 and 24 V ranges was almost similar. The optimal fog harvesting design merges hierarchical geometry from multilayer to single-layer designs. Electrowetting offers a promising avenue for active fog harvesting systems in diverse regions, though field testing under natural wind and humidity gradients is essential. Further research may explore low-power electrowetting configurations (e.g., pulsed voltages) and environmentally resilient coatings (hydrophobic) to ensure long-term functionality. Computational modeling of droplet dynamics on electrified meshes could refine design parameters for real-world deployment. The integration of advanced mesh geometries with electrowetting technology holds significant potential to enhance fog harvesting efficiency. However, transitioning from laboratory-scale success to field applications demands interdisciplinary innovation in materials science, energy systems, and environmental engineering.

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AUTHOR CONTRIBUTIONS

G.S.Y.: conceptualization, formal analysis and writing-original draft; **D.J.:** supervision and corrections; **A.M.**: writing – review and editing, **S.S.** and **L.L.D.**: performed experiments on electrowetting; **V.S.**: assisted in

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electrowetting experiment, **B.C.**: interpreting the data, development, and debriefing; **G.T.P.**: revisions and editing. All authors of this paper have directly participated in this study's writing, editing, planning, analysis, and execution.

DECLARATION OF COMPETING INTEREST

The authors declare there is no competing interest of financial and personal relationships with other people or organizations in this research work.

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DATA AVAILABILITY STATEMENT

Data cannot be made publicly available; readers should contact the corresponding author for details.

CONFLICT OF INTEREST

The authors declare there is no conflict.

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